FairHindiSER: Problem Statement & Motivation

Fair Speech Emotion Recognition for Bilingual Hindi–English Audio

What is SER?

Speech Emotion Recognition (SER) classifies emotional state from raw audio for classes such as *Angry*, *Happy*, *Neutral*, and *Sad*.

Where is it used?

- Mental health monitoring
- Call-centre analytics
- Human–robot interaction

Foundation model:

wav2vec 2.0 – 7-layer CNN + 12 Transformer layers, pre-trained on English audiobooks.

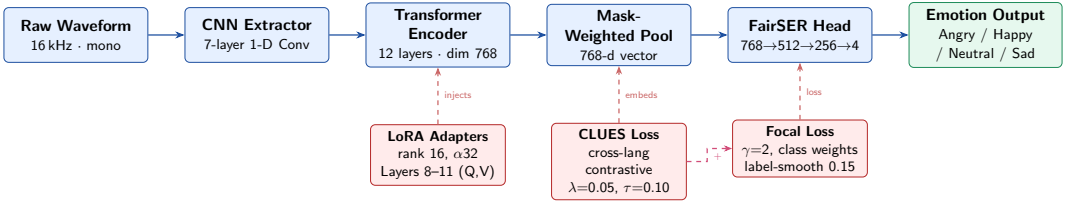
Two Critical Biases

Language	English-centric training misreads Hindi prosody
Gender	Male-skewed corpora create unequal performance

Our goal:

- Balanced bilingual dataset (EN + HI)
- Fair five-stage fine-tuning pipeline
- AudioTrust-based evaluation

System Architecture



Training Summary

Stage	Key change	Macro F1	Lang gap	Gender gap
1. Head-only	Train classifier head only; frozen backbone baseline	0.5506	0.0800	0.0855
2. LoRA + Focal	Add LoRA on top layers and Focal Loss for imbalance	0.6294	0.0536	0.0291
3. CLUES	Add cross-language contrastive debiasing	0.6515	0.0263	0.0098
4. Unfreezing	Progressive encoder unfreezing with Optuna-tuned learning rates	**0.7400**	**0.0008**	**0.0060**

Takeaway

- Stage 2 gives the biggest gain
- Stage 3 narrows language bias
- Stage 4 gives the best fairness/accuracy trade-off

Key settings kept

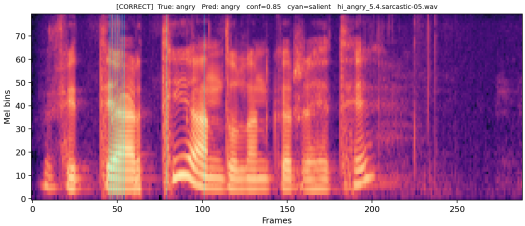
- LoRA on Q/V, layers 8–11
- Focal Loss with $\gamma = 2$
- CLUES contrastive debiasing

Final outcome

- English F1: 0.7167
- Hindi F1: 0.8117
- Balanced multilingual SER

AudioTrust Evaluation, Four Pillars

Dimension	Score	
Fairness	0.877	**Strong**
Robustness	0.765	Good
Explainability	0.910	**Excellent**
Privacy	0.595	Moderate
Overall	**0.787**	



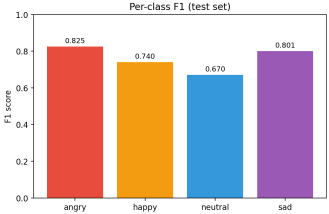
Integrated Gradients saliency map from the report: attribution mass concentrates on voiced speech regions.

Highlights

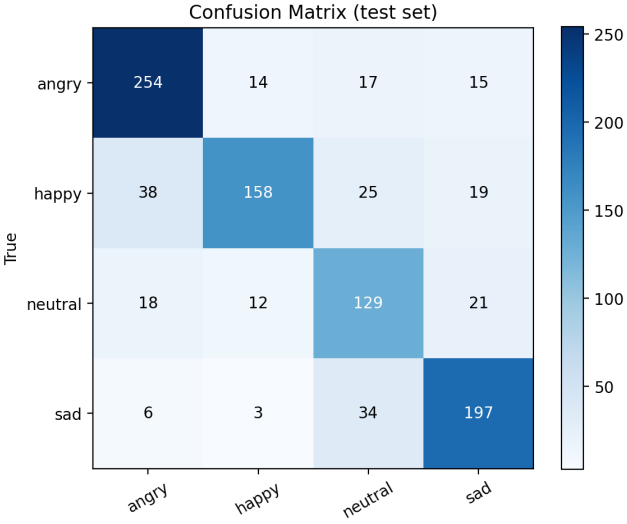
Fairness: γ_{gender}=0.898, γ_{lang}=0.901
Robustness: mean drop 0.067, weakest at 10 dB noise
Privacy: MIA AUC 0.590, confidence gap 0.037

Per-Class Performance

Emotion	F1	Note
Angry	0.825	Clear cues
Happy	0.740	Overlap
Neutral	0.670	Hardest
Sad	0.801	Good separation
Macro F1	**0.761**	



Neutral remains the hardest class.



Conclusion

What we achieved

- **Macro F1 = 0.761** on a balanced 6,400-sample corpus
- Gender gap reduced from 0.0855 to **0.0060**
- Language gap reduced to **0.0008**
- AudioTrust score: **0.786**

What worked best

- LoRA for efficient adaptation
- CLUES for cross-language fairness
- Progressive unfreezing with Optuna

Main Limitation

10 dB SNR noise lowers F1 to 0.46. A denoising front-end is needed before deployment.

Future Work

- Spectral denoising front-end
- More languages: Tamil, Bengali
- Streaming / real-time inference

FairHindiSER shows that fairness-aware staged tuning reduces demographic gaps without sacrificing quality.